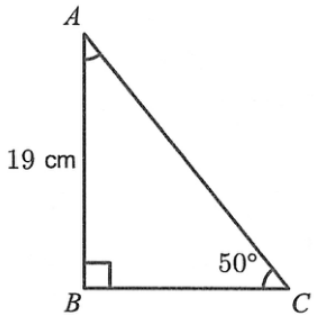
Trigonometry

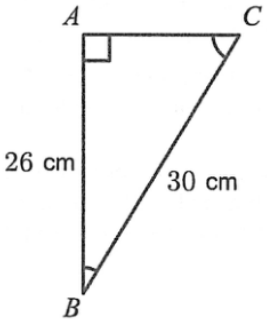
Questions 1–140

1.39 — Question 1. Find all the missing angles and sides in the following right-angled triangles, giving your answers to three significant figures.

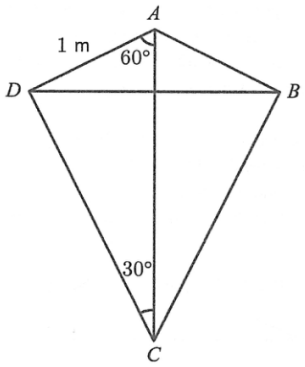
a)



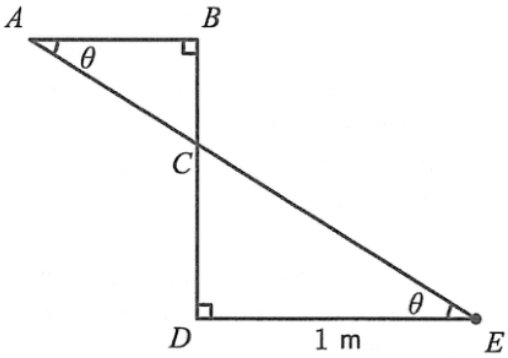
b)



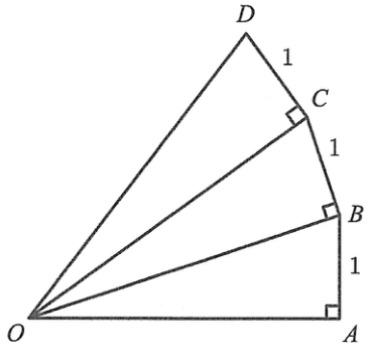
1.39 — Question 2. The diagram below shows the kite $ABCD$, where angle $DAC = 60°$ and angle $ACD = 30°$. It is also given that the length of AD is 1 m. Find the exact area of the kite.



1.39 — Question 3. In the diagram below, the angles ABC and CDE are both 90°, and the length DE is 1 m. It is given that BCD is a straight line. The area of triangle ABC is one third of the area of triangle CFE. Find the exact length of AB.

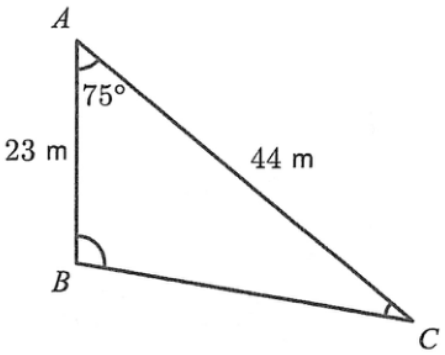


1.39 — Question 4. In the diagram below, the angles OAB, OBC and OCD are all 90°, and the lengths AB, BC and CD are all 1 unit.



Find the length OA given that $\frac{OD}{OA} = \sqrt{2}$.

1.39 — Question 5. Find the area of the triangle shown in the diagram below, giving your answer to three significant figures.



1.39 — Question 6. It is given that ABC is a triangle where angle ABC is $45°$, side $AB = 20$ cm, and side $BC = 15$ cm. Find the length of side AC.

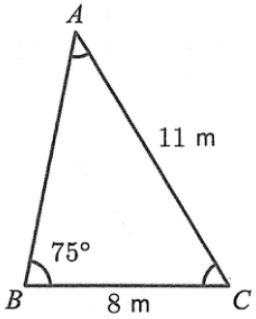
1.39 — Question 7. It is given that ABC is a triangle where AB is 9 cm, BC is 12 cm, and angle $ACB = 40°$.

a) Use the sine rule to find two possible values for angle BAC, giving your answers to one decimal place.

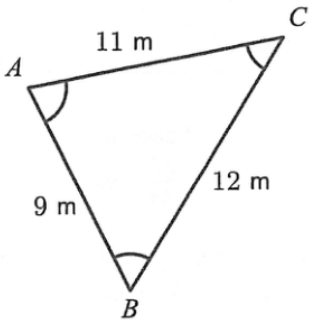
b) Hence draw and label two possible triangles ABC.

1.39 — Question 8. Find all the missing angles and sides in the following triangles, giving your answers to three significant figures.

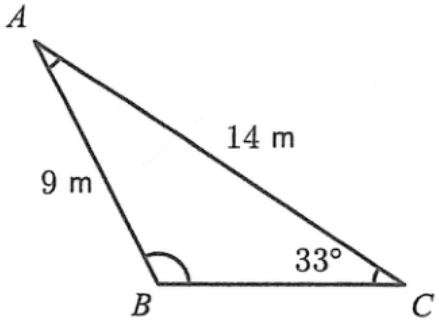
a)



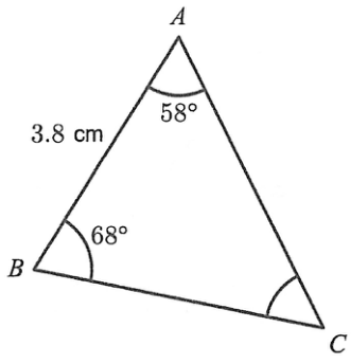
b)



c)

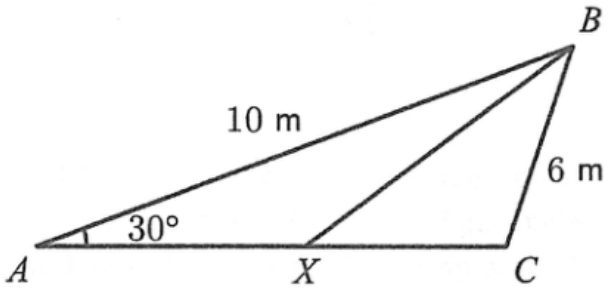


1.39 — Question 9. Find the area of the triangle shown in the diagram below, giving your answer to three significant figures.



1.39 — Question 10. It is given that ABC is a triangle where AB is 16 cm, AC is 9 cm and angle BAC is $x°$. The area of the triangle is 25 cm^2. Find the two possible values of x to one decimal place.

1.39 — Question 11. The diagram below shows the triangle ABC. It is given that angle ACB is obtuse. Find the length of XC given that BX bisects the triangle.



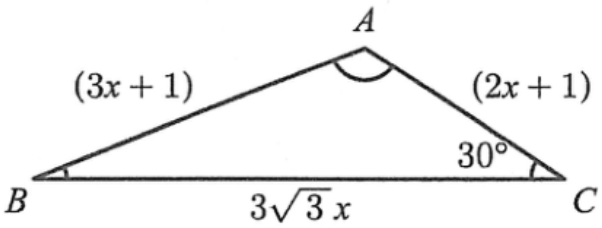
1.39 — Question 12. It is given that ABC is a triangle where AB is $5m$ cm, AC is $2m^2$ cm and angle BAC is $45°$. The area of the triangle is 640 cm^2. Show that $m = k\sqrt{2}$ where k is an integer to be found.

1.39 — Question 13. The diagram below shows the triangle ABC.

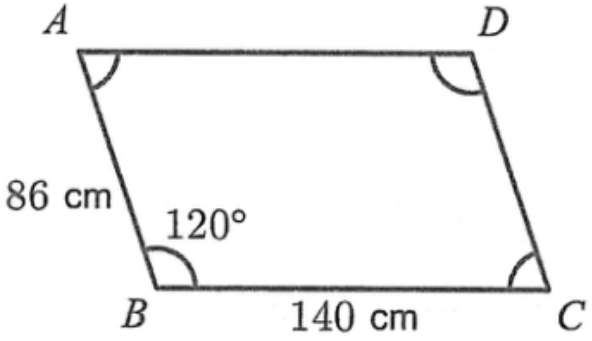
a) Show that $11x - 4x^2 = 0$

b) Find the value of x

c) Find the largest angle in the triangle.

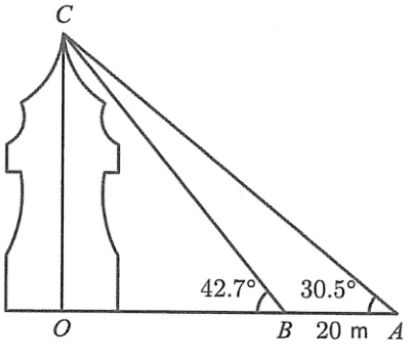


1.39 — Question 14. The diagram below shows a parallelogram with sides of length 86 cm and 140 cm. It is given that angle $ABC = 120°$. Show that the exact area of the parallelogram is $k\sqrt{3}$ m^2 where k is a rational number.



1.39 — Question 15. It is given that a triangle has sides of length b, $b+1$ and $b+2$, where $b > 0$. The largest angle in the triangle is labelled x. Use the cosine rule to show that $\cos x = \frac{b-3}{2b}$.

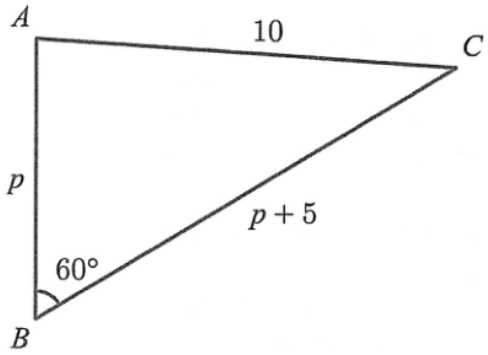
1.39 — Question 16. Ben measures the angle of elevation from where he is standing to the top of a tall building to be 30.5°. He then walks straight towards the building for 20 metres, stops, and measures the angle of elevation again. This time it is 42.7°. Determine the height of the building.



1.39 — Question 17. Three archaeological digs, A, B and C, are drawn on a map. B is 180 metres due South of A. C is 320 metres from A on a bearing of 075°. Find

a) the bearing of B from A
b) the bearing of A from C
c) the bearing of B from C

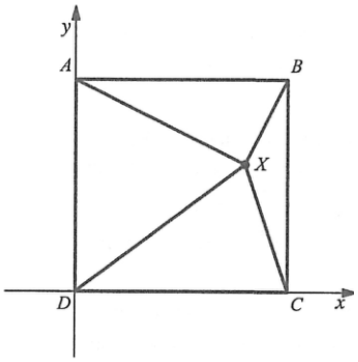
1.39 — Question 18. The diagram below shows a triangle, ABC, with sides of length p, $p+5$ and 10. $\angle ABC = 60°$. Find the exact value of p.



1.39 — Question 19. The area of a regular pentagon is 60 cm^2. Calculate the length of one of its sides.

1.39 — Question 20. The area of a regular octagon is 450 cm^2. Calculate the length of one of its sides.

1.39 — Question 21. The diagram below shows a square, $ABCD$, with area 100 units^2 drawn on the coordinate axes. The square has been divided up into four triangles, XBC, XAB, XCD, and XAD, with areas 10 units^2, 20 units^2, 30 units^2, and 40 units^2 respectively.



a) X has exact coordinates (p, q). Find the value of p and the value of q
b) Y has exact coordinates (r, q), where $0 < r < p$. The triangle XYD has an area that is at least half of the area of triangle XAD. Find the range of possible values of r

1.40 — Question 1. Sketch the following graphs for $0° \leq x \leq 360°$:
a) $y = \sin x$
b) $y = \cos x$
c) $y = \tan x$

1.40 — Question 2. Solve the following equations:

a) $\sin\theta = 0$, $0° \leq \theta < 360°$

b) $\sin\theta = 1$, $0° \leq \theta < 360°$

c) $\sin\theta = -1$, $0° \leq \theta < 360°$

1.40 — Question 3. Solve the following equations:

a) $\cos\theta = 0$, $0° \leq \theta < 360°$

b) $\cos\theta = 1$, $0° \leq \theta < 360°$

c) $\cos\theta = -1$, $0° \leq \theta < 360°$

1.40 — Question 4. Solve the following equations:

a) $\tan\theta = 0$, $0° \leq \theta < 360°$

b) $\tan\theta = 1$, $0° \leq \theta < 360°$

c) $\tan\theta = -1$, $0° \leq \theta < 360°$

1.40 — Question 5. Find the values of x for which $y = \sin x$ is an increasing function, for $0° \leq x \leq 360°$.

1.40 — Question 6. Find the values of x for which $y = \cos x$ is a decreasing function, for $0° \leq x \leq 360°$.

1.40 — Question 7. Solve the following equations:

a) $\sin\theta = \frac{1}{3}$, $0° \leq \theta < 360°$

b) $\cos\theta = \frac{2}{3}$, $0° \leq \theta < 360°$

c) $\tan\theta = \frac{4}{3}$, $0° \leq \theta < 360°$

1.40 — Question 8. Solve the following equations:

a) $\sin\theta = 0.4$, $-360° \leq \theta < 0°$

b) $\cos\theta = \frac{3}{7}$, $-360° \leq \theta < 0°$

c) $\tan\theta = \frac{4}{9}$, $-360° \leq \theta < 0°$

1.40 — Question 9. Solve the following equations:

a) $\sin\theta = -0.2$, $0° \leq \theta < 360°$

b) $\cos\theta = -\frac{4}{7}$, $0° \leq \theta < 360°$

c) $\tan\theta = -4$, $0° \leq \theta < 360°$

1.40 — Question 10. Solve the following equations:

a) $\sin\theta = -0.9$, $-360° \leq \theta < 0°$
b) $\cos\theta = -\frac{3}{5}$, $-360° \leq \theta < 0°$
c) $\tan\theta = -\frac{9}{2}$, $-360° \leq \theta < 0°$

1.40 — Question 11. Solve the following equations:

a) $\sin\theta = \frac{3}{8}$, $-360° \leq \theta < 360°$
b) $\cos\theta = 0.7$, $-360° \leq \theta < 360°$
c) $\tan\theta = 6$, $-360° \leq \theta < 360°$
d) $\sin\theta = -\frac{6}{7}$, $-360° \leq \theta < 360°$
e) $\cos\theta = -\frac{2}{9}$, $-360° \leq \theta < 360°$
f) $\tan\theta = -1.3$, $-360° \leq \theta < 360°$

1.40 — Question 12. Find the values of x for which $y = \sin x$ is a decreasing function, for $-720° \leq x \leq -360°$.

1.40 — Question 13. Find the values of x for which $y = \cos x$ is an increasing function, for $180° \leq x \leq 540°$.

1.40 — Question 14. Solve the following equations:

a) $\sin\theta = \frac{4}{7}$, $0° \leq \theta < 720°$

b) $\cos\theta = -\frac{8}{9}$, $720° \leq \theta < 1080°$

c) $\tan\theta = -9$, $-720° \leq \theta < -540°$

1.40 — Question 15. Solve the following equations:

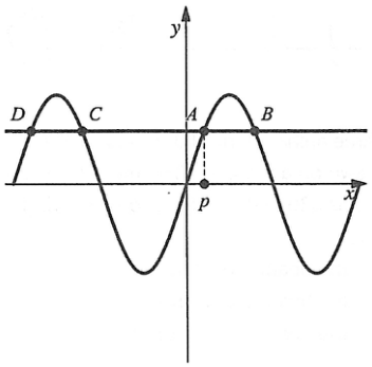
a) $\frac{1}{\sin\theta} = 18$ for $0° \leq \theta < 180°$

b) $\frac{5}{\tan\theta} = 13$ for $180° \leq \theta < 540°$

c) $\frac{0.1}{\cos\theta} = 2$ for $-180° \leq \theta < 180°$

1.40 — Question 16. A student is solving a problem with a triangle. They correctly use the sine rule to find $\frac{\sin A}{18} = \frac{\sin 20^\circ}{15}$. Find the two possible values of A.

1.40 — Question 17. The graph below shows the curve $y = \sin x$ for $-360^\circ \leq x \leq 360^\circ$, and the line $y = k$ where $0 < k < 1$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.

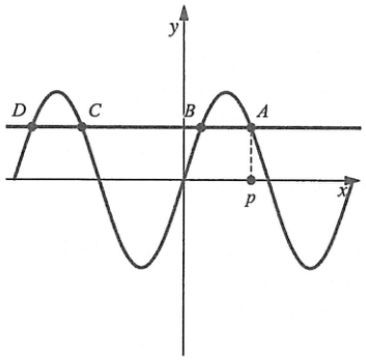


1.40 — Question 18. It is given that $\sin 60° = \frac{\sqrt{3}}{2}$. Explain why $\sin 480° = \frac{\sqrt{3}}{2}$.

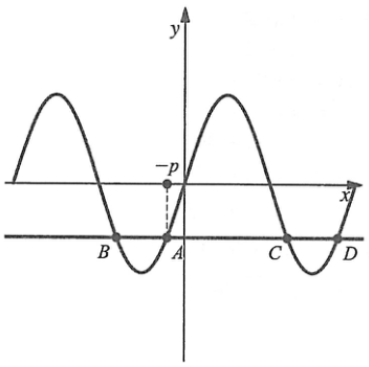
1.40 — Question 19. It is given that $\cos 45° = \frac{\sqrt{2}}{2}$. Explain why $\cos 495° = -\frac{\sqrt{2}}{2}$.

1.40 — Question 20. It is given that $\tan(-210°) = -\frac{\sqrt{3}}{3}$. Explain why $\tan 390° = \frac{\sqrt{3}}{3}$.

1.40 — Question 21. The graph below shows the curve $y = \sin x$ for $-360° \leq x \leq 360°$, and the line $y = k$ where $0 < k < 1$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



1.40 — Question 22. The graph below shows the curve $y = \sin x$ for $-360° \leq x \leq 360°$, and the line $y = k$ where $-1 < k < 0$. The point A has x-coordinate $-p$, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



1.40 — Question 23. Solve the following equations:

a) $\cos^2\theta = \frac{1}{16}$ for $0° \leq \theta < 360°$

b) $3\tan^2\theta = 0.8$ for $-180° \leq \theta < 180°$

c) $10\sin^2\theta = 7$ for $0° \leq \theta < 540°$

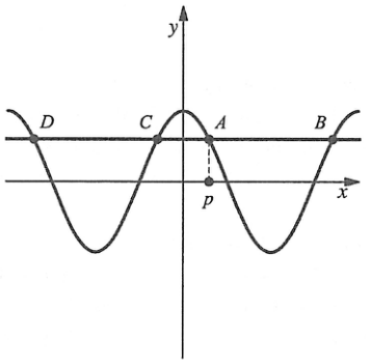
1.40 — Question 24. Solve the following equations:

a) $\cos^2\theta = 2\cos\theta$ for $0° \leq \theta < 360°$

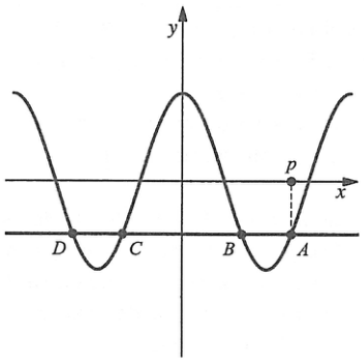
b) $7\tan^2\theta + 4\tan\theta = 0$ for $0° \leq \theta < 360°$

c) $2\sin\theta = 5\sin^2\theta$ for $0° \leq \theta < 360°$

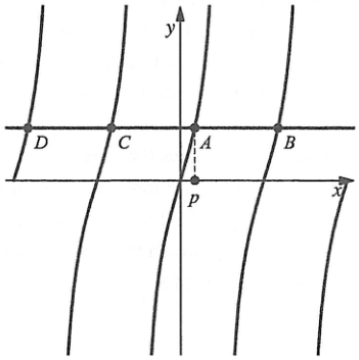
1.40 — Question 25. The graph below shows the curve $y = \cos x$ for $-360° \leq x \leq 360°$, and the line $y = k$ where $0 < k < 1$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



1.40 — Question 26. The graph below shows the curve $y = \cos x$ for $-360° \leq x \leq 360°$, and the line $y = k$ where $-1 < k < 0$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



1.40 — Question 27. The graph below shows the curve $y = \tan x$ for $-360° \leq x \leq 360°$, and the line $y = k$ where $0 < k < 1$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



1.40 — Question 28. Solve the following equations:

a) $3\sin\theta\cos\theta = \cos\theta$ for $0° \leq \theta < 360°$

b) $8\sin\theta\tan\theta - 9\sin\theta = 0$ for $0° \leq \theta < 360°$

c) $6\cos\theta - 5\sin\theta\cos\theta = 0$ for $0° \leq \theta < 360°$

1.40 — Question 29. Solve the following equations:

a) $2\sin^2\theta + \sin\theta - 1 = 0$ for $0° \leq \theta < 360°$

b) $3\cos^2\theta - \cos\theta - 2 = 0$ for $0° \leq \theta < 360°$

c) $4\tan^2\theta + 5\tan\theta - 6 = 0$ for $0° \leq \theta < 360°$

1.40 — Question 30. Solve the equations for $0° \leq \theta < 360°$:

a) $6\sin\theta = 1 + \frac{1}{\sin\theta}$

b) $9\cos\theta = \frac{16}{\cos\theta} - 32$

c) $30\tan\theta = 23 + \frac{40}{\tan\theta}$

1.40 — Question 31. Prerequisite knowledge for Q31: Quadratics It is given that $f(x) = 3x^2 - 12x + 17$.

a) Write $f(x)$ in the form $a(x+b)^2 + c$, where a, b and c are integers.

b) Write down the minimum point of the curve with equation $y = f(x)$.

It is given that $g(\theta) = 3\sin^2\theta - 12\sin\theta + 17$.

c) Using your answers to a) and b), find the minimum value of $g(\theta)$ and then the smallest positive value of θ, in degrees, for which this minimum occurs.

1.40 — Question 32. Write down the period of the following functions in degrees:

a) $y = \sin x$
b) $y = \cos x$
c) $y = \tan x$
d) $y = \sin 2x$
e) $y = \cos 3x$
f) $y = \tan 4x$
g) $y = \sin \frac{1}{3}x$
h) $y = \cos \frac{2}{3}x$
i) $y = \tan \frac{4}{5}x$

1.40 — Question 33. State the number of solutions to the following equations. You are not required to solve them.

a) $\sin 2x = 0.3$ for $0° \leq x < 360°$
b) $\sin 2x = 0.3$ for $0° \leq x < 180°$
c) $\sin 2x = 0.3$ for $0° \leq x < 720°$

1.40 — Question 34. State the number of solutions to the following equations. You are not required to solve them.

a) $\cos 2x = 0.8$ for $0° \leq x < 180°$
b) $\cos 3x = 0.8$ for $0° \leq x < 360°$
c) $\cos 4x = 0.8$ for $0° \leq x < 720°$

1.40 — Question 35. State the number of solutions to the following equations. You are not required to solve them.

a) $\tan 2x = 0.8$ for $0° \leq x < 180°$
b) $\tan 3x = 0.8$ for $0° \leq x < 720°$
c) $\tan 4x = 0.8$ for $-270° \leq x < 360°$

1.40 — Question 36. Write down the period of the following functions in degrees:

a) $y = \sin(x - 15°)$
b) $y = \cos(x + 80°)$
c) $y = \tan(x - 300°)$
d) $y = \sin(60° - x)$
e) $y = \cos(45° - x)$
f) $y = \tan(5° - x)$
g) $y = \sin(2x + 50°)$
h) $y = \cos(4x - 90°)$
i) $y = \tan(9x + 35°)$
j) $y = \sin(100° - \frac{1}{5}x)$
k) $y = \cos(\frac{315° - x}{8})$
l) $y = \tan(\frac{28° - 5x}{9})$

1.40 — Question 37. Solve the following equations:

a) $\sin(\theta + 10°) = 0.3$ for $0° \leq \theta < 360°$
b) $\sin(\theta - 50°) = \frac{2}{11}$ for $0° \leq \theta < 360°$
c) $\sin(\theta + 205°) = -0.4$ for $0° \leq \theta < 360°$

1.40 — Question 38. Solve the following equations:

a) $\cos(\theta - 60°) = \frac{4}{11}$ for $0° \leq \theta < 360°$

b) $\cos(\theta + 25°) = 0.85$ for $0° \leq \theta < 360°$

c) $\cos(\theta - 195°) = -\frac{7}{12}$ for $0° \leq \theta < 360°$

1.40 — Question 39. Solve the following equations:

a) $\tan(\theta - 45°) = 2.5$ for $0° \leq \theta < 360°$

b) $\tan(\theta + 80°) = \frac{5}{6}$ for $0° \leq \theta < 360°$

c) $\tan(\theta - 215°) = -4.2$ for $0° \leq \theta < 360°$

1.40 — Question 40. Find the values of x for which $y = \sin(x - 30°)$ is an increasing function, for $0° \leq x \leq 360°$.

1.40 — Question 41. Solve the following equations:

a) $\sin 2\theta = \frac{\sqrt{2}}{2}$ for $0° \leq \theta < 360°$

b) $\cos 2\theta = \frac{\sqrt{2}}{4}$ for $0° \leq \theta < 360°$

c) $\tan 2\theta = \frac{\sqrt{3}}{3}$ for $0° \leq \theta < 360°$

1.40 — Question 42. Solve the following equations:

a) $\sin 3\theta = 0.8$ for $0° \leq \theta < 360°$

b) $\cos 3\theta = -0.3$ for $0° \leq \theta < 360°$

c) $\tan 3\theta = -3$ for $0° \leq \theta < 360°$

1.40 — Question 43. Solve the following equations:

a) $\sin \frac{1}{2}\theta = 0.2$ for $0° \leq \theta < 720°$

b) $\cos \frac{1}{4}\theta = -\frac{4}{5}$ for $0° \leq \theta < 3600°$

c) $\tan \frac{2}{3}\theta = -5$ for $0° \leq \theta < 270°$

1.40 — Question 44. Solve the following equations:

a) $\sin(2\theta - 25°) = \frac{1}{2}$ for $0° \leq \theta < 360°$

b) $\cos(3\theta + 50°) = \frac{1}{2}$ for $0° \leq \theta < 360°$

c) $\tan(2\theta + 65°) = 1$ for $0° \leq \theta < 360°$

1.40 — Question 45. Solve the following equations:

a) $\sin(4\theta + 161°) = \frac{5}{6}$ for $0° \leq \theta < 180°$

b) $\cos(6\theta - 94°) = 0.03$ for $-90° \leq \theta < 90°$

c) $\tan(5\theta - 213°) = 8$ for $100° \leq \theta < 200°$

1.40 — Question 46. Find the values of x for which $y = \cos 2x$ is a decreasing function, for $0° \leq x \leq 180°$.

1.40 — Question 47. Find the values of x for which $y = \sin 3x$ is an increasing function, for $0° \leq x \leq 360°$.

1.40 — Question 48. Solve the following equations:

a) $\sin(84° - 3\theta) = \frac{1}{2}$ for $0° \leq \theta < 360°$

b) $\cos(115° - 2\theta) = \frac{1}{2}$ for $0° \leq \theta < 360°$

c) $\tan(5° - 4\theta) = 1$ for $0° \leq \theta < 360°$

1.40 — Question 49. Solve the following equations:

a) $\sin(28° - 5\theta) = 0.1$ for $0° \leq \theta < 144°$

b) $\cos(77° - 3\theta) = 0.99$ for $0° \leq \theta < 300°$

c) $\tan(174° - 8\theta) = -0.56$ for $0° \leq \theta < 55°$

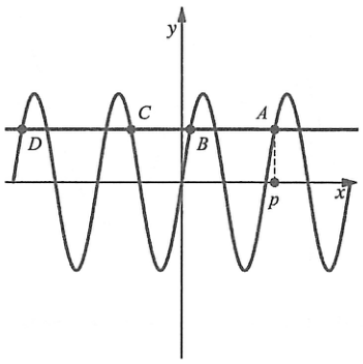
1.40 — Question 50. Solve the following equations:

a) $\sin^2 2\theta - \frac{1}{4} = 0$ for $0° \leq \theta < 360°$

b) $\cos^2 3\theta - \frac{1}{16} = 0$ for $0° \leq \theta < 180°$

c) $\tan^2 5\theta - 121 = 0$ for $-40° \leq \theta < 40°$

1.40 — Question 51. The graph below shows the curve $y = \sin 2x$ for $-360° \leq x \leq 360°$, and the line $y = k$ where $0 < k < 1$. The point A has x-coordinate p. Write down the x-coordinates of B, C and D in terms of p.



1.40 — Question 52. Solve the following equations:

a) $\sin(\theta - 40°) = \sin 80°$ for $0° \leq \theta < 360°$

b) $\sin(\theta + 35°) = \sin 280°$ for $0° \leq \theta < 360°$

c) $\cos(\theta + 15°) = \cos 50°$ for $0° \leq \theta < 360°$

d) $\tan(\theta - 70°) = \tan 20°$ for $0° \leq \theta < 360°$

1.40 — Question 53. Solve the following equations:

a) $\sin(35° - \theta) = \sin 75°$ for $0° \leq \theta < 360°$

b) $\cos(2\theta + 80°) = \cos 55°$ for $0° \leq \theta < 360°$

c) $\tan(20° - 3\theta) = \tan 140°$ for $0° \leq \theta < 360°$

1.40 — Question 54. Prerequisite knowledge for Q54: Polynomials It is given that $f(x) = 45x^3 - 63x^2 - 2x + 8$.

a) Show that $(5x - 2)$ is a factor of $f(x)$

b) Hence solve the equation $45\cos^3\theta - 63\cos^2\theta - 2\cos\theta + 8 = 0$ for $0° \leq \theta \leq 360°$.

1.42 — Question 1. Write each of the following expressions in terms of $\sin x$:

a) $\sin(x + 180°)$

b) $\sin(-x)$

c) $\sin(x + 360°)$

d) $\sin(x - 180°)$

e) $\sin(x - 360°)$

f) $\sin(180° - x)$

g) $\sin(360° - x)$

h) $\sin(540° - x)$

i) $\sin(x - 540°)$

j) $\sin(x + 540°)$

k) $\sin(x + 720°)$

l) $\sin(900° - x)$

m) $\sin(3600° - x)$

1.42 — Question 2. Write each of the following expressions in terms of $\cos x$:

a) $\cos(x - 180°)$
b) $\cos(-x)$
c) $\cos(180° - x)$
d) $\cos(x + 180°)$
e) $\cos(x + 360°)$
f) $\cos(360° - x)$
g) $\cos(x - 360°)$
h) $\cos(540° - x)$
i) $\cos(x + 540°)$
j) $\cos(x - 540°)$
k) $\cos(x - 720°)$
l) $\cos(900° - x)$
m) $\cos(3600° - x)$

1.42 — Question 3. Write each of the following expressions in terms of $\tan x$:

a) $\tan(x + 180°)$
b) $\tan(-x)$
c) $\tan(x + 360°)$
d) $\tan(x - 180°)$
e) $\tan(360° - x)$
f) $\tan(180° - x)$
g) $\tan(x - 360°)$
h) $\tan(540° - x)$
i) $\tan(x - 540°)$
j) $\tan(x + 540°)$
k) $\tan(x + 720°)$
l) $\tan(900° - x)$
m) $\tan(3600° - x)$

1.42 — Question 4. Write each of the following expressions in terms of $\cos x$:

a) $\sin(x + 90°)$
b) $\sin(x - 90°)$
c) $\sin(x + 270°)$
d) $\sin(90° - x)$
e) $\sin(270° - x)$
f) $\sin(x + 450°)$
g) $\sin(x - 450°)$
h) $\sin(450° - x)$
i) $\sin(x - 630°)$
j) $\sin(630° - x)$
k) $\sin(x + 810°)$
l) $\sin(990° - x)$
m) $\sin(9090° - x)$

1.42 — Question 5. Write each of the following expressions in terms of $\sin x$:

a) $\cos(x + 90°)$
b) $\cos(x - 90°)$
c) $\cos(x + 270°)$
d) $\cos(90° - x)$
e) $\cos(270° - x)$
f) $\cos(x + 450°)$
g) $\cos(x - 450°)$
h) $\cos(450° - x)$
i) $\cos(x - 630°)$
j) $\cos(630° - x)$
k) $\cos(x + 810°)$
l) $\cos(990° - x)$
m) $\cos(9450° - x)$

1.42 — Question 6. It is given that $\sin x = \frac{1}{3}$ and x is acute. Find the exact values of:

a) $\cos x$

b) $\tan x$

1.42 — Question 7. It is given that $\cos x = \frac{2}{7}$ and x is acute. Find the exact values of:

a) $\sin x$

b) $\tan x$

1.42 — Question 8. It is given that $\tan x = \frac{3}{10}$ and x is acute. Find the exact values of:

a) $\sin x$

b) $\cos x$

1.42 — Question 9. It is given that $\sin x = \frac{2}{9}$ and x is obtuse. Find the exact values of:

a) $\cos x$

b) $\tan x$

1.42 — Question 10. It is given that $\cos x = -\frac{3}{7}$ and x is obtuse. Find the exact values of:

a) $\sin x$

b) $\tan x$

1.42 — Question 11. It is given that $\tan x = -\frac{11}{20}$ and x is obtuse. Find the exact values of:

a) $\sin x$

b) $\cos x$

1.42 — Question 12. It is given that $\sin x = -\frac{5}{6}$ and $180° < x < 270°$. Find the exact values of:

a) $\cos x$

b) $\tan x$

1.42 — Question 13. It is given that $\cos x = -\frac{1}{3}$ and $180° < x < 270°$. Find the exact values of:

a) $\sin x$

b) $\tan x$

1.42 — Question 14. It is given that $\tan x = \frac{7}{2}$ and $180° < x < 270°$. Find the exact values of:

a) $\sin x$

b) $\cos x$

1.42 — Question 15. It is given that $\sin x = -\frac{1}{10}$ and $270° < x < 360°$. Find the exact values of:

a) $\cos x$

b) $\tan x$

1.42 — Question 16. It is given that $\cos x = \frac{14}{15}$ and $270° < x < 360°$. Find the exact values of:

a) $\sin x$

b) $\tan x$

1.42 — Question 17. It is given that $\tan x = -\frac{20}{3}$ and $270° < x < 360°$. Find the exact values of:

a) $\sin x$

b) $\cos x$

1.42 — Question 18. It is given that $\cos 75° = \frac{\sqrt{2-\sqrt{3}}}{2}$.

a) Find the value of $\sin 75°$

b) Using your answer to a), show that $\tan 75° = \sqrt{7+4\sqrt{3}}$

1.42 — Question 19. Simplify the following trigonometric expressions as far as possible, writing your answers in terms of $\sin x$ and $\cos x$ only:

a) $\cos x \tan x$

b) $\sin x \tan x$

c) $\frac{\tan x}{\sin x}$

d) $\frac{\tan x}{\cos x}$

1.42 — Question 20. Simplify the following trigonometric expressions as far as possible, writing your answers in terms of $\sin x$ and $\cos x$ only:

a) $\sin x \cos x \tan x$
b) $\sin x \cos^2 x \tan^2 x$
c) $\frac{\sin x}{\cos x \tan^2 x}$
d) $\frac{\tan^2 x}{\sin x \cos x}$

1.42 — Question 21. Simplify the following trigonometric expressions:

a) $1 - \cos^2 x$
b) $\sqrt{1 - \sin^2 x}$
c) $9 \cos^2 x - 9$
d) $8 \cos^2 x + 8 \sin^2 x$

1.42 — Question 22. Simplify the following trigonometric expressions:

a) $\frac{1-\sin^2 x}{\cos^2 x}$

b) $\frac{1-\cos^2 x}{\sin^2 x}$

c) $\frac{4-4\cos^2 x}{5-5\sin^2 x}$

d) $\frac{\tan x}{6-\frac{6}{\cos^2 x}}$

1.42 — Question 23. Solve the following equations:

a) $\sin\theta = \cos\theta$, $0° \leq \theta < 360°$

b) $2\sin\theta = 3\cos\theta$, $0° \leq \theta < 360°$

c) $8\sin\theta + 5\cos\theta = 0$, $0° \leq \theta < 360°$

d) $7\cos\theta - 2\sin\theta = 0$, $0° \leq \theta < 360°$

1.42 — Question 24.

a) Sketch the graphs of $y = \cos x$ and $y = 2\sin x$ on the same axes between $0°$ and $360°$.

b) Determine the coordinates of where the two graphs intersect.

1.42 — Question 25. Solve the following equations:

a) $6\sin 3\theta = 5\cos 3\theta$, $0° \leq \theta < 240°$

b) $7\sin \frac{1}{2}\theta - 9\cos \frac{1}{2}\theta = 0$, $0° \leq \theta < 720°$

c) $11\cos \frac{1}{3}\theta - 25\sin \frac{1}{3}\theta = 0$, $0° \leq \theta < 2160°$

1.42 — Question 26. Solve the following equations:

a) $4\tan\theta + \cos\theta = 0,\ 0° \leq \theta < 360°$

b) $\sin\theta\tan\theta + 6 = 0,\ 0° \leq \theta < 360°$

c) $8\tan 2\theta + \cos 2\theta = 0,\ 0° \leq \theta < 360°$

1.42 — Question 27. Solve the following equations:

a) $2\cos^2\theta = 3\sin\theta,\ 0° \leq \theta < 180°$

b) $4\sin^2\theta = 15\cos\theta,\ 0° \leq \theta < 540°$

c) $7\cos^2\theta = 48\sin\theta,\ 0° \leq \theta < 360°$

d) $10\sin^2\theta = 99\cos\theta,\ -180° \leq \theta < 360°$

1.42 — Question 28. Write the expression $2\cos^4 x + \cos^2 x$ in terms of $\sin x$ only.

1.42 — Question 29. Solve the following equations:

a) $2\cos^2\theta + 3\sin\theta - 3 = 0$, $0° \leq \theta < 360°$

b) $3\sin^2\theta - 5\cos\theta - 5 = 0$, $0° \leq \theta < 360°$

c) $6\cos^2\theta + \sin\theta - 4 = 0$, $0° \leq \theta < 360°$

d) $20\sin^2\theta - 7\cos\theta - 14 = 0$, $0° \leq \theta < 360°$

1.42 — Question 30. Solve the following equations:

a) $3\sin^2\theta + \sin\theta - 3\cos^2\theta + 2 = 0,\ 0° \leq \theta < 360°$

b) $4\cos^2\theta - 16\cos\theta + 3\sin^2\theta - 19 = 0,\ 0° \leq \theta < 360°$

c) $10\sin^2\theta - 11\cos^2\theta + 46\sin\theta + 35 = 0,\ 0° \leq \theta < 360°$

1.42 — Question 31.

a) Solve the equation $5\cos^2 x + 3\cos x + 2 = 4\sin^2 x$ for $-360° \leq x < 360°$

b) Hence find the smallest positive solution to the equation $5\cos^2(2\theta + 50°) + 3\cos(2\theta + 50°) + 2 = 4\sin^2(2\theta + 50°)$

1.42 — Question 32. Solve $2\sin^2 x + 3 = \tan 5x - 2\cos^2 x$ for $0° \leq x < 72°$.

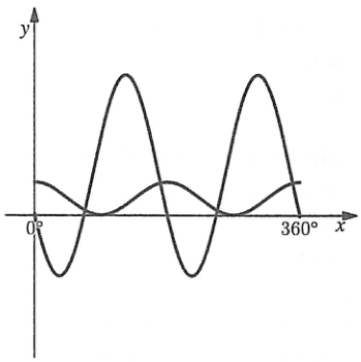
1.42 — Question 33. Solve the following equations:

a) $10\sin\theta\tan\theta + 21 = 0$, $0° \leq \theta < 360°$

b) $2\sin\theta\tan\theta + 20\cos\theta + 15 = 0$, $0° \leq \theta < 360°$

c) $\frac{2\cos\theta}{\tan\theta} = 18\sin\theta - 3$, $0° \leq \theta < 360°$

1.42 — Question 34. The graph below shows the curves with equation $y = 12\sin^2 x - 28\sin x \cos x$ and $y = 5\cos^2 x$ from $0°$ to $360°$. Determine the coordinates of where the two graphs intersect.



1.42 — Question 35. Solve the following equations:

a) $\sin\theta\tan\theta + 4\cos\theta = 5\sin\theta$, $0° \leq \theta < 360°$

b) $5\sin\theta\tan\theta = 3\cos\theta + 14\sin\theta$, $0° \leq \theta < 360°$

c) $10\sin\theta\tan\theta + 3\sin\theta = 18\cos\theta$, $0° \leq \theta < 360°$

1.42 — Question 36. Solve the following equations:

a) $2\cos\theta + 13\cos\theta\tan\theta + 6\sin\theta\tan\theta = 0,\ 0° \leq \theta < 360°$

b) $15\cos\theta + 6\sin\theta\tan\theta = 19\cos\theta\tan\theta,\ 0° \leq \theta < 360°$

1.42 — Question 37. Prove the following trigonometric identities:

a) $(\sin x + \cos x)^2 \equiv 1 + 2\sin x\cos x$

b) $\sin^3 x + \cos^3 x \equiv (\sin x + \cos x)(1 - \sin x\cos x)$

c) $1 + \tan^2 x \equiv \frac{1}{\cos^2 x}$

d) $\frac{1}{\tan^2 x} + 1 \equiv \frac{1}{\sin^2 x}$

1.42 — Question 38. Prove the following trigonometric identities:

a) $\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} \equiv \frac{1}{\sin x \cos x}$

b) $\frac{1}{\cos x+1} - \frac{1}{\cos x-1} \equiv \frac{2}{\sin^2 x}$

c) $\frac{\cos x}{1-\sin x} + \frac{\cos x}{1+\sin x} \equiv \frac{2}{\cos x}$

d) $\frac{\cos x}{\sin x-1} + \frac{\cos x}{\sin x+1} \equiv -2\tan x$

1.42 — Question 39. Prove the following trigonometric identities:

a) $\frac{1}{1+\tan^2 x} \equiv \cos^2 x$

b) $\frac{\tan x}{1+\tan^2 x} \equiv \sin x \cos x$

c) $\frac{1}{1-\tan^2 x} \equiv \frac{\cos^2 x}{1-2\sin^2 x}$

d) $\frac{\cos^2 x}{1-\tan^2 x} \equiv \frac{1}{1-\tan^4 x}$

1.42 — Question 40. Prove the following trigonometric identities:

a) $\frac{1}{\frac{1}{\sin^2 x}-1} \equiv \tan^2 x$

b) $\frac{1}{\frac{1}{\cos^2 x}-1} \equiv \frac{1}{\sin^2 x} - 1$

c) $\frac{1}{\frac{1}{\tan^2 x}+1} \equiv \sin^2 x$

d) $\frac{1}{\frac{1}{\sin^2 x}+\frac{1}{\cos^2 x}} \equiv (\sin x \cos x)^2$

1.42 — Question 41. Prove the following trigonometric identities:

a) $\sin^4 x - \cos^4 x \equiv \sin^2 x - \cos^2 x$

b) $\sin^2 x + \frac{\sin^4 x}{\cos^2 x} \equiv \tan^2 x$

c) $\cos^4 x + \sin^2 x + \sin^2 x \cos^2 x + \sin^2 x \tan^2 x \equiv \cos^2 x + \tan^2 x$

d) $\sin^2 x \cos^2 x - 1 + \sin^4 x - \tan^2 x \equiv \sin^2 x - \frac{1}{\cos^2 x}$

1.42 — Question 42. Prove the following trigonometric identities:

a) $\frac{\sin x}{1-\tan x} + \frac{\cos x}{1-\frac{1}{\tan x}} \equiv 0$

b) $\frac{\cos x}{1-\tan x} + \frac{\sin x}{1-\frac{1}{\tan x}} \equiv \sin x + \cos x$

c) $\frac{\sin x}{1-\tan x} - \frac{\cos x}{1-\frac{1}{\tan x}} \equiv \frac{2\sin x - 2\cos x}{2-\frac{1}{\sin x \cos x}}$

d) $\frac{\cos x}{1-\tan x} - \frac{\sin x}{1-\frac{1}{\tan x}} \equiv \frac{1}{\cos x - \sin x}$

1.42 — Question 43. Prove the following trigonometric identities:

a) $\frac{\sin x}{1-\cos x} \equiv \frac{1+\cos x}{\sin x}$

b) $\frac{\cos x}{1-\sin x} \equiv \frac{1+\sin x}{\cos x}$

c) $\frac{\sin x}{1-\tan x} \equiv \frac{\sin x \cos x}{\cos x - \sin x}$

d) $\frac{1-\sin x}{1-\cos x} \equiv \frac{1+\cos x}{\sin^2 x} - \frac{1+\cos x}{\sin x}$

1.42 — Question 44. Prove the following trigonometric identities:

a) $\frac{\tan x+\frac{1}{\tan x}}{\frac{1}{\cos x}+\frac{1}{\sin x}} \equiv \frac{1}{\sin x+\cos x}$

b) $\frac{\cos x-\frac{1}{\cos x}}{\frac{1}{\sin x}-\frac{1}{\cos x}} \equiv \frac{\sin^2 x}{1-\frac{1}{\tan x}}$

c) $\frac{\sin x-\frac{1}{\sin x}}{\frac{1}{\sin x}-\frac{1}{\cos x}} \equiv \frac{\cos^3 x}{\sin x-\cos x}$

1.42 — Question 45. Prove the following trigonometric identities:

a) $\frac{\sin x\cos x}{1+\sin x+\cos x-\cos^2 x} \equiv \frac{1}{\frac{1}{\sin x}+\frac{1}{\cos x}+\tan x}$

b) $\frac{\sin x\cos^2 x}{\cos^3 x-2\cos^2 x-\cos x+1} \equiv \frac{1}{\frac{\tan x}{\cos x}-\frac{1}{\sin x}-\tan x}$

1.42 — Question 46. Prove the following trigonometric identities:

a) $(\frac{1}{\cos x} - \sin x)(\frac{1}{\sin x} - \cos x) \equiv \frac{(\sin x \cos x - 1)^2}{\sin x \cos x}$

b) $(\frac{1}{\sin x} - \sin x)(\frac{1}{\cos x} - \cos x) \equiv \frac{1}{\tan x + \frac{1}{\tan x}}$

1.42 — Question 47. Prove the following trigonometric identity: $\frac{2\cos^2 x - 1}{\cos x - \sin x} \equiv \frac{1}{\sin x - \cos x + \frac{2\cos^2 x}{\sin x + \cos x}}$.

1.42 — Question 48. Prove the following trigonometric identity: $\sin^2 A + \sin^2 B - 1 \equiv \sin^2 A \sin^2 B - \cos^2 A \cos^2 B$.

1.42 — Question 49. Isla is attempting to solve the equation $\sqrt{2}\cos^2 x + \sin x - \sqrt{2} = 0$ for $180° < x < 360°$. Her attempt is shown below:

$$\begin{aligned}
&\sqrt{2}\cos^2 x + \sin x + \sqrt{2} = 0 \\
\Rightarrow\ &\sqrt{2}(\sin^2 x - 1) + \sin x + \sqrt{2} = 0 \\
\Rightarrow\ &\sqrt{2}\sin^2 x - \sqrt{2} + \sin x + \sqrt{2} = 0 \\
\Rightarrow\ &\sqrt{2}\sin^2 x + \sin x = 0 \\
\Rightarrow\ &\sqrt{2}\sin x + 1 = 0 \\
\Rightarrow\ &\sin x = -\frac{1}{\sqrt{2}} \\
\Rightarrow\ &x = -45°
\end{aligned}$$

Explain the two errors that Isla has made.

1.42 — Question 50.

a) Show that the equation $20\sin x\tan x + 31\sin x = 7\cos x$ can be written in the form $(4\tan x + 7)(5\tan x - 1) = 0$

b) Hence solve the equation $20\sin 4\theta\tan 4\theta + 31\sin 4\theta = 7\cos 4\theta$ for $0° \leq \theta < 45°$

1.42 — Question 51.

a) Show that $\frac{\tan x}{\frac{1}{\cos x} - \cos x} \equiv \frac{1}{\sin x}$

b) Using your answer to a), solve the equation $\frac{\tan x}{\frac{1}{\cos x} - \cos x} = 6$ for $0° \leq x < 360°$

c) Hence solve the equation $\frac{\tan(2\theta - 55°)}{\frac{1}{\cos(2\theta - 55°)} - \cos(2\theta - 55°)} = 6$ for $0° \leq \theta < 180°$

1.42 — Question 52.

a) Show that $\frac{\cos^3 x}{\sin x} + \sin x \cos x \equiv \frac{1}{\tan x}$

b) Using your answer to a), solve the equation $\frac{\cos^3 x}{\sin x} + \sin x \cos x = \frac{1}{20}$ for $0° \leq x < 360°$

c) Hence solve the equation $\frac{\cos^3(2\theta+9°)}{\sin(2\theta+9°)} + \sin(2\theta + 9°)\cos(2\theta + 9°) = \frac{1}{20}$ for $0° \leq \theta < 180°$

1.42 — Question 53. Solve the equation $\sin^4\theta - \cos^4\theta = \frac{1}{2}$ for $0° \leq \theta \leq 360°$.

1.42 — Question 54. Solve the equation $2\cos^4\theta + \cos^2\theta\sin\theta - 2\cos^2\theta = 0$ for $0° \leq \theta \leq 360°$.

1.43 — Question 1. State the period of each of the following functions:

a) $f(x) = \sin 2x$

b) $f(x) = \cos 6x$

c) $f(x) = \tan x$

d) $f(x) = \sin\left(\frac{1}{2}x\right)$

e) $f(x) = \cos\left(\frac{1}{4}x\right)$

f) $f(x) = \tan(2x - 45)$

g) $f(x) = \sin\left(\frac{360}{7}x\right)$

h) $f(x) = \cos\left(\frac{90}{13}x + \frac{360}{13}\right)$

i) $f(x) = \sin\left(\frac{72}{73}(x - 30)\right)$

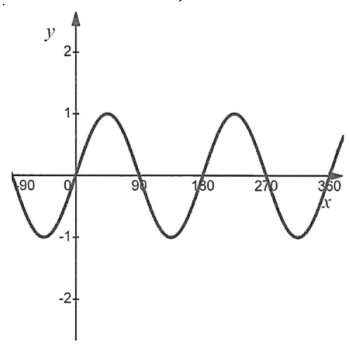
j) $f(x) = \cos(15(x + 7))$

1.43 — Question 2. In each of the following cases, the curve

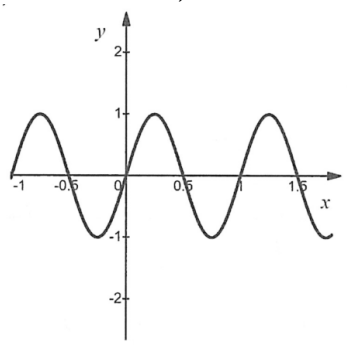
$$y = \sin x$$

has been stretched parallel to the x-axis. State the equation of each curve.

a)



b)

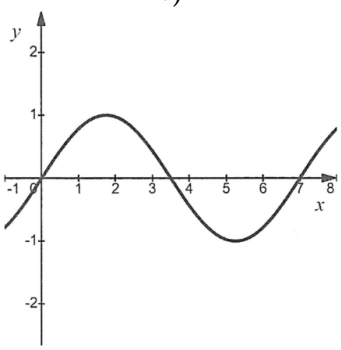


1.43 — Question 2 (continued). In each of the following cases, the curve

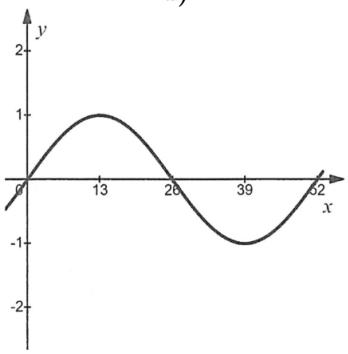
$$y = \sin x$$

has been stretched parallel to the x-axis. State the equation of each curve.

c)



d)

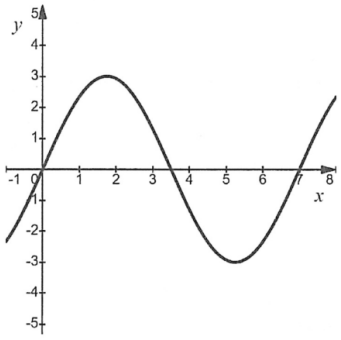


1.43 — Question 3. All the following graphs are transformations of

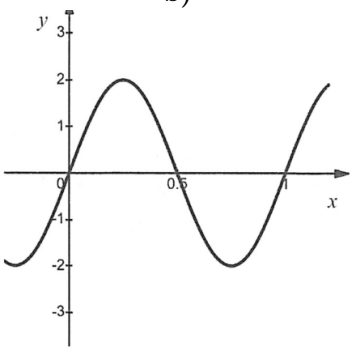
$$y = \sin x.$$

State the equation of each:

a)



b)

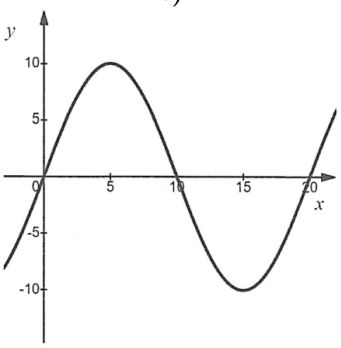


1.43 — Question 3 (continued). All the following graphs are transformations of

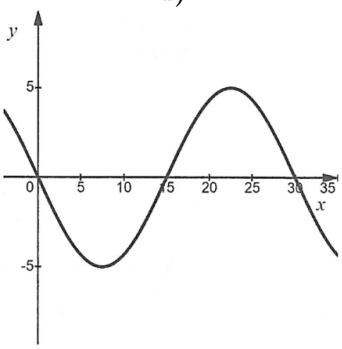
$$y = \sin x.$$

State the equation of each:

c)



d)



1.43 — Question 4. State the amplitude of each of the following periodic functions:

a) $y = \sin 2x$

b) $y = 2 + \cos(15x + 3)$

c) $y = 40 + 18\sin(12x - 4)$

d) $y = 3 - 5\sin(0.5x + 1.1)$

1.43 — Question 5. For each of the following functions state the minimum and maximum of the function:

a) $y = \sin 3x$

b) $y = 4 + \cos(12x^\circ - 7)$

c) $y = 25 + 15\sin(30x + 2)$

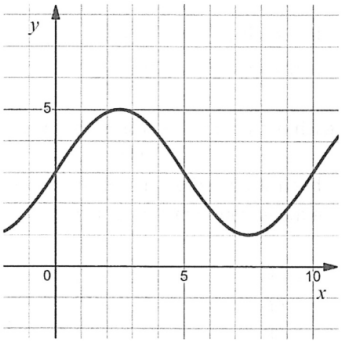
d) $y = 3 - 8\sin(0.5x + 0.2)$

1.43 — Question 6. All the following graphs are of the form

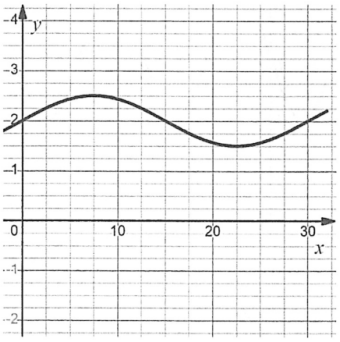
$$y = a + b\sin(cx).$$

In each case, find the equation of the graph:

a)



b)

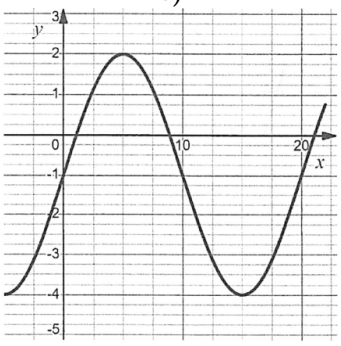


1.43 — Question 6 (continued). All the following graphs are of the form

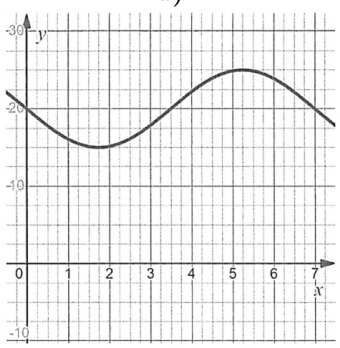
$$y = a + b\sin(cx).$$

In each case, find the equation of the graph:

c)



d)



1.43 — Question 7. The time of sunrise, y hours since midnight for each day x of a year may be modelled by the equation

$$y = a + b\cos x^\circ.$$

It is known that the latest sunrise is at 8:00 am on January 1st and the earliest is at 4:30 am, six months later.

a) Find the values of a and b.

b) Suggest how the model may be improved.

1.43 — Question 8. A person's blood pressure, y, oscillates between 120 and 80 with their heartbeat occurring every 1 second.

a) Find a model in the form

$$y = a + b\sin kt^\circ$$

where t is the time that has passed in seconds and a, b, k are constants.

b) Explain why this model will not always be accurate for predicting blood pressure.

1.43 — Question 9. The size of a population, P, of wild mice is recorded over time, and may be modelled by a formula of the form

$$P = 20 + 6\sin(3t - 10)^\circ, \qquad 0 \leq t \leq 121$$

where t is the number of days since the start of the year. According to the model, find

a) the number of mice in the population after 5 days have passed.

b) the maximum and minimum number of mice in the population.

c) the days on which the maximum and the minimum number of mice in the population occurs.

1.43 — Question 10. The profits on the sales of ice cream for a shop, £P, may be given by the equation

$$P = 55 - 50\cos\left(\frac{72}{73}(x - 45)\right)^\circ$$

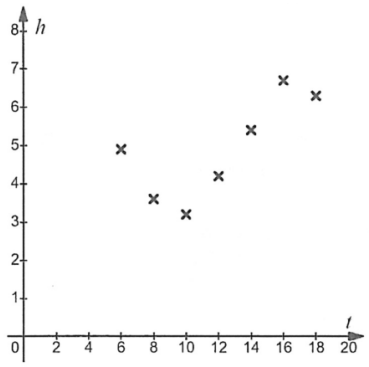
where x is the number of days since the 1st January. According to the model,

a) write down the difference between the largest and smallest values of the profits.

During the 136th day of the year, the rate of change of the profits is at its maximum.

b) Find the next day when the model predicts the rate of change will be the same.

1.43 — Question 11. Laiyla is monitoring the height of the tide, h metres, in her local harbour. At various points of time, she records the height along with the time t hours after midnight.



1.43 — Question 11 (continued). Laiyla models the data with the equation

$$h = 5 + 1.7\sin(25t - 330)^\circ$$

where t is the number of hours since midnight.

a) Find the minimum height of the tide predicted by Laiyla's model.

b) Find the maximum number of consecutive hours that Laiyla's model predicts that the height of the tide will exceed 6 metres.

c) Find the second time after midnight at which the tide has a height of 6 metres.

1.43 — Question 11 (continued). Further along the coast, a second location's tide has the same times for its high and low tides, varying in depth by 3.3 metres. At high tide, it is 2.55 metres less deep than the high tide found by Laiyla's model.

d) Make appropriate refinements to Laiyla's equation to create a model for the height of the tide.